

Deep Learning for Time Series LSST Classification

Adapting Foundation Models for Astronomical Data

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Outline

- 1 Context & Motivation
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- **The Large Synoptic Survey Telescope (LSST):**
 - Observatory located in Chile.
 - Goal: Understand the changing sky, discover and measure millions of celestial objects.
- **The challenge:**
 - Classifying simulated light curves from the PLAsTiCC challenge.
 - Light curves measure the brightness of celestial objects over time.
- **Objective:**
 - Establish robust baselines.
 - Adapt and evaluate time-series **Foundation Models** for astronomical classification.

Dataset characteristics

Data Structure

- Multivariate time-series.
- **36 time steps** per sample.
- **6 passbands** (u, g, r, i, z, y) as features.

Splits & Classes

- Train: 2,459 samples.
- Test: 2,466 samples.
- **14 distinct classes** of astronomical events.
- Highly **unbalanced** class distribution.

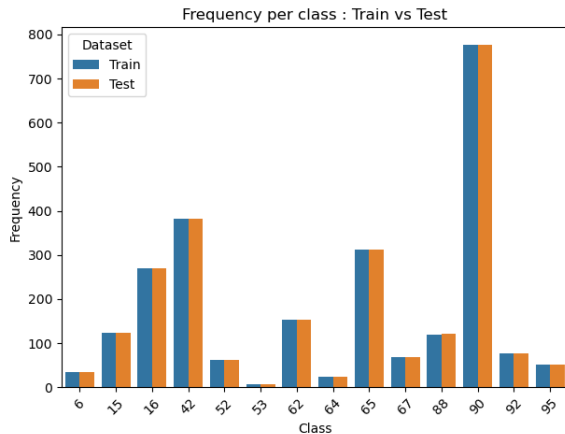


Figure: Unbalanced LSST dataset

Evaluation metrics

Due to the lack of domain priors and the severe class imbalance, we must evaluate our models carefully:

Macro F1 Score (Primary metric)

- Calculates the metric independently for each class before taking the unweighted mean.
- Ensures minority classes are preserved and not overshadowed by dominant classes.

Weighted Logarithmic Loss (W-LogLoss)

- Aligns with the original PLAsTiCC benchmark.
- Heavily penalizes incorrect, high-confidence probability predictions.
- Evaluates the model's confidence and predictive uncertainty.

$$L = -\frac{1}{M} \sum_{j=1}^M \left(\frac{\sum_{i=1}^{N_j} v_{ij} \ln(p_{ij})}{\sum_{i=1}^{N_j} v_{ij}} \right) \quad (1)$$

Establishing Baselines

We implemented three classical deep learning architectures with a balanced class weighting scheme in the Cross-Entropy loss:

- **1D-CNN:**
 - Captures local temporal variations across the 6 passbands.
 - Uses max pooling and adaptive average pooling.
- **1D-RNN:**
 - Bidirectional RNN with Dropout regularization.
- **Bidirectional LSTM:**
 - Concatenates the final forward and backward hidden states for prediction.

Foundation Models (1/3): Mantis

- Time-series foundation model using a Vision Transformer layer.
- Generates embeddings per channel, concatenated for multivariate input.
- Evaluated using *Linear Probing (LP)* and Fine-tuning.

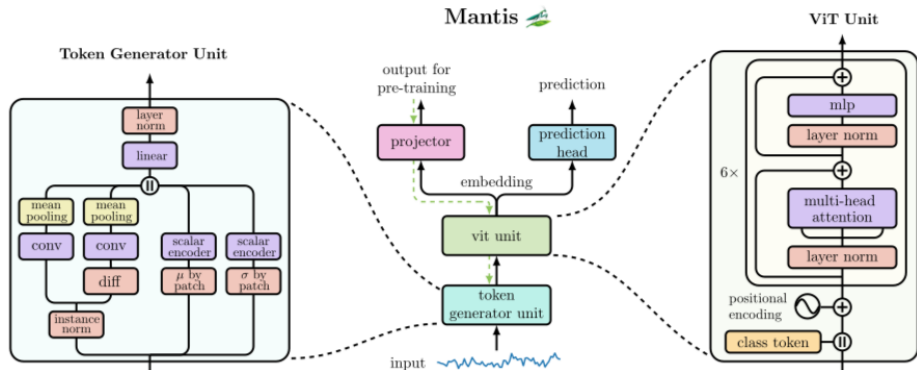


Figure: Mantis architecture

Foundation Models (2/3): MultiRocket

- Fast, efficient classification method without GPU pretraining.
- Used 1,000 & 6,250 random convolutional kernels.
- Extracts Proportion of Positive Values (PPV) and max values.
- Paired with Logistic Regression.

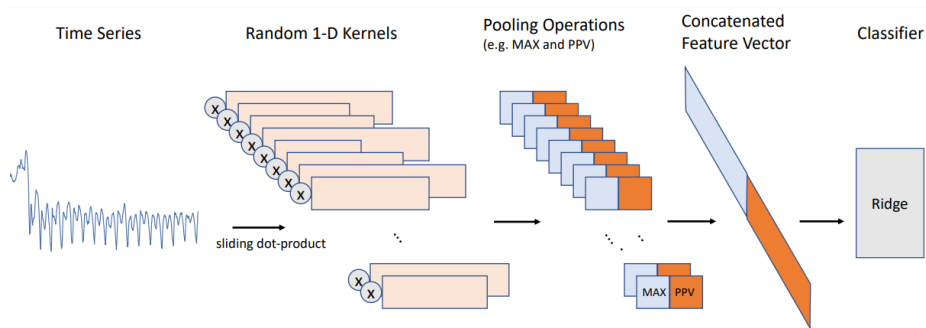
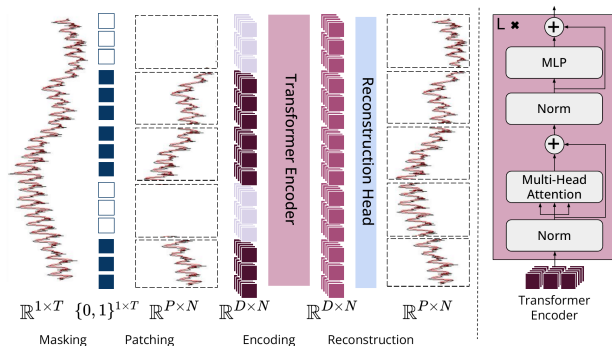


Figure: MultiRocket architecture

Foundation Models (3/3): MOMENT-FM

- A 341M parameter foundation model (24 transformer blocks).
- Replaced default linear head with a **2-layer MLP** to increase capacity for the 14 LSST classes.
- Padded sequences from $T = 36$ to $T = 40$ (multiple of patch size 8) using attention masks.



Training Strategy:

- ① **Warmup (15 epochs):** Train only the new MLP head; freeze backbone.
- ② **Fine-Tuning:** Unfreeze last 4 blocks. Use differential learning rates (3×10^{-4} for head, 5×10^{-5} for backbone) with cosine annealing.

Experimental Setup

- **Data splits:**
 - Stratified 80/20 train/validation split
 - Final test set: 2,466 samples.
- **Preprocessing:**
 - Instance-wise z-score normalization.
 - Applied also to validation and test sets.
- **Hyperparameter tuning:**
 - Selected based on Validation Macro F1.
 - Best configurations retrained on combined Train+Validation set.

Results on Test Set

Model	Macro F1 $\uparrow$	W-LogLoss $\downarrow$
1D-CNN	0.3014	2.0490
1D-RNN	0.1788	3.9173
LSTM	0.4205	1.3965
Mantis V1 + LP	0.4672	3.1756
Mantis V2 + LP	0.4778	1.9473
MultiRocket (1000)	0.5095	1.6681
MultiRocket (6250)	0.5278	1.9026
MOMENT-FM	0.4043	3.9449

MultiRocket achieves the highest Macro F1 score, while LSTM achieves the best LogLoss calibration among all models.

- **Baselines:** Bidirectional LSTM vastly outperforms standard RNN/CNN by successfully capturing structural profiles across time.
- **Mantis vs. MultiRocket:**
 - Mantis yields good F1 scores but suffers from overconfident misclassifications (high W-LogLoss).
 - **MultiRocket dominates** by capturing diverse temporal patterns effectively without immense computational overhead.
- **Why did MOMENT-FM underperform?**
 - *Sequence Length:* Pretrained on long sequences; $T = 36$ yields only 5 patches (too short for deep transformers).
 - *Data Scarcity:* 1,967 samples are insufficient to adapt a 341M parameter model, even with layer freezing.

Key Takeaway

For short, small-scale astronomical time-series datasets like the PLAsTiCC challenge, **highly optimized classical methods (MultiRocket)** currently outperform massive, deep foundation models (MOMENT-FM).

Our complete pipeline is available on GitHub:
https://github.com/antoineverier/lst_classification

Thank you for your attention.